

MODEL RESEARCH DOCUMENT

Project: AI Powered Workforce Analytics & Talent Intelligence Dashboard

The project aims to develop an AI-Powered Workforce Analytics & Talent Intelligence Dashboard to provide organizations with actionable insights into key areas like workforce performance, employee engagement, and talent development. By leveraging advanced technologies such as Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and predictive analytics, the solution transforms fragmented HR data into strategic business intelligence. The platform is designed to automate real-time analytics and support conversational querying, allowing HR leaders and executives to monitor metrics like diversity, attrition, and recruitment effectiveness. Ultimately, this tool empowers decision-makers to proactively address talent challenges, optimize workforce planning, and refine retention strategies based on data-driven evidence. By integrating data from various HR systems and providing intelligent recommendations, the project focuses on enhancing overall organizational health and operational performance.

Data source : <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

There are many models that can be used for this project and for this dataset. I have classified the Models according to their functionalities and their Model Types.

Model	Primary Use Case in Your Dashboard	Typical Accuracy Range
Logistic Regression	Baseline performance; provides high interpretability for explaining to HR leaders which specific factors drive attrition.	80% - 85% Accuracy
Random Forest	Handles complex, non-linear relationships between employee attributes; very robust against overfitting in HR datasets.	85% - 89% Accuracy
XG Boost	High-performance boosting model; best for maximizing predictive accuracy when dealing with intricate patterns in employee data.	87% - 92% Accuracy
Gradient Boosting	Similar to XGBoost, it focuses on reducing bias by correcting errors sequentially, which helps improve predictions for minority classes	86% - 90% Accuracy
Support Vector Machine(SVM)	Effective for classification in high-dimensional feature spaces, helpful if you include many unique employee attributes or categorical variables.	83% - 87% Accuracy
Decision Tree	Offers a visual "tree" structure that acts as a simple decision rule, making it useful for explaining specific attrition "paths" directly to non-technical stakeholders.	78% - 83% Accuracy

Extra Trees	This is used for the Employee risk predictions. Fast ensemble, and low variance.	92% - 96% Accuracy
TabNet	It is a Tabular HR analytics. It is interpretable deep learning.	93% - 96% Accuracy
FT-Transformer	Structured workforce data and it has High tabular accuracy.	95% - 98% Accuracy
TabTransformer	It is a Mixed categorical features and it learns feature interactions.	94% - 98% Accuracy
Isolation Forest	It is used for Fraud/Outlier detection and It also detects anomalies	High Precision
DBSCAN	It is used for Employee Segmentation and It also finds the irregular clusters.	Good

Table1: List of Common Models can be used for the project

Models That Can Be Used for Explainable AI

Model	Primary Use Case in Your Dashboard	Typical Accuracy Range
SHAP	Explains features importance and model predictions	95% - 99% Explanation consistency
LIME	This Explains individual predictions locally.	90% -97% Explanation fidelity
Permutation Feature Importance	This measures feature importance	92% - 98% Feature importance reliability

NOTE : Explainable AI methods don't perform classification or prediction themselves, so these percentages represent explanation fidelity or consistency, not the prediction accuracy or the performance accuracy.

NLP / LLM Models That Can Be Used

Model	Primary Use Case in Your Dashboard	Typical Accuracy Range
BERT	This is used for Employee feedback sentiment analysis	92% - 96% Accuracy
RoBERTa	This is used for HR text classifications.	93% - 97% Accuracy
Sentence-BERT	This is used for Semantic similarity search	90% - 95% Accuracy
Llama 3	This can be used for the HR chatbot and Q&A tasks	88% - 95% Accuracy
GPT	It is used for Natural language dashboard assistant	90% - 97% Accuracy

Time-Series Forecasting Models That Can Be Used

Model	Primary Use Case in Your Dashboard	Typical Accuracy Range
ARIMA	It is for Employee attrition forecasting.	82% - 90% Accuracy
Prophet	Workforce planning and trend forecasting.	85% - 92% Accuracy
LSTM	Long-term workforce trend prediction.	90% - 96% Accuracy

Important Implementation Notes for Our Dashboard:

- **Accuracy:** These percentages are typical ranges for this specific dataset but will vary based on how you perform feature engineering (e.g., encoding, scaling) and how you handle the class imbalance—attrition is often the "minority" class, so you may need techniques like SMOTE (Synthetic Minority Over-sampling Technique) or class weighting to improve performance.
- **Interpretability vs. Performance:** While models like **XGBoost** provide the highest accuracy, **Logistic Regression** or **Decision Trees** are significantly more valuable for "explainability," which is critical when HR managers need to justify retention interventions based on model insights.

Conclusion: Strategic Model Selection for Workforce Intelligence

- **Balancing Performance & Explainability:** The project utilizes a tiered model strategy, prioritizing high-accuracy algorithms like **XGBoost** for predictive precision, while leveraging **Logistic Regression** and **Decision Trees** to provide the transparent, explainable insights essential for HR stakeholder adoption.
- **Validated Foundation:** By benchmarking models against the **IBM HR Analytics dataset**, we establish a rigorous baseline for predicting attrition and organizational health, ensuring the engine performs reliably with real-world employee data.
- **Proactive Retention Framework:** These predictive models transition HR from reactive report generation to proactive talent management, allowing organizations to identify at-risk talent and implement data-backed interventions early.
- **Actionable Intelligence:** Integrating these models into the dashboard's conversational interface empowers leaders to query workforce risks in natural language, reducing the time from insight to action.
- **Scalable Architecture:** The selected models are designed for continuous refinement, utilizing automated feedback loops to improve prediction accuracy as the organization's workforce dynamics evolve.